Indian Statistical Institute
Second Semester Examinations: 2025-26

Course Name: M. Math, 2^{nd} year
Subject Name : PDE
Maximum Marks: 60, Duration: Three hours
Date: 25.04.2026, 2:30 PM – 5:30 PM

- You may use any results proved in class. Any other results require proof.
- You will be awarded the full marks for a question if you justify using correct arguments that there is a mistake in the question.
- Let $B \subset \mathbb{R}^n$, $n \geq 2$ be the open unit ball, centered at the origin.

1. **[10 marks]** Let T be any isometry of $\mathbb{R}^n$, with $n \geq 2$, that is T is surjective with $\|Tx - Ty\| = \|x - y\|$, for each $x, y \in \mathbb{R}^n$. Then show that there is a non-constant bounded solution (pointwise/classical sense) of

$$\partial_t u = \Delta_x u \text{ in } (0, \infty) \times \mathbb{R}^n,$$

satisfying $u(t, Ty) = u(t, y)$, for all $y \in \mathbb{R}^n$, $t > 0$. (Hint: Choose appropriate boundary value $u(0, \cdot)$.)

2. Consider the PDE

$$(-\Delta)^p u = f, \tag{1}$$

where $(-\Delta)^p$ is the p-fold composition of the Laplacian operator.

(a) **[7 marks]** For $V = \mathbb{R}^{10}$, and any $f \in L^1(V) \cap L^2(V)$, find the possible values of $p \in \mathbb{N}$ for which (1) has a distributional solution $u \in L^2(V)$.

(b) **[8 marks]** For $V = B$, and any $f \in C^\infty(V) \cap L^2(V)$, show that for any $p \in \mathbb{N}$, (1) has a pointwise/classical solution in $C^\infty(V)$.

3. Consider for $w \in H_0^1(B)$, the PDE

$$\Delta u = \cos(\|\nabla w\|). \tag{2}$$

(a) **[7 marks]** Show that a unique weak solution u of (2) exists in $H_0^1(B)$, for each $w \in H_0^1(B)$.

(b) **[5 marks]** Denote the weak solution u by $T(w)$ and show that the nonlinear map

$$T : H_0^1(B) \to H_0^1(B)$$

is continuous, that is

$$w_n \to w \in H_0^1(B) \implies u_n - u = T(w_n) \to T(w) \in H_0^1(B).$$

(Hint: Use $x \mapsto \cos(x)$ is Lipschitz, the energy estimate for solutions for the function $u_n - u$, and PI.)

(c) **[5 marks]** Show that T has range contained in a bounded subset of $H^2(B) \cap H_0^1(B)$ (bounded in the $H^2(B)$-norm).
(Hint: Use the fact that $\cos \in L^\infty$, energy estimate for solutions for the function $u = T(w)$, and PI.)

(d) **[3 marks]** Show that

$$\Delta u = \cos(\|\nabla u\|)$$

has a solution in $H^2(B) \cap H_0^1(B)$.

4. **[7 marks]** Find an explicit integral representation formula for a solution $u : (0, \infty) \times \mathbb{R}^n \to \mathbb{R}$ of the problem

$$\partial_t u - \Delta_x u - \langle b, \nabla_x u \rangle = f, \quad u(0, \cdot) \equiv 0$$

where $f \in C^2_1((0, \infty) \times \mathbb{R}^n)$ has compact support, and $b \in \mathbb{R}^n$ is a vector. (Hint: Consider the function $v(t, x) = u(t, x - tb)$ first.)

5. **[8 marks]** Let $I = (-1, 1)$. Show that $u : I \to \mathbb{R}$, defined

$$u(x) = 1 - \|x\| \quad \text{for } x \in I,$$

is contained in $H^1(I)$, but not in $H^2(I)$.